C# Machine Problem: International Cargo Freight & Customs Clearance System

Problem Scenario

A global freight forwarding and logistics company requires a console-based application to evaluate cargo packages, calculate shipping costs, determine customs tariffs, and issue an operational clearance decision.

Your task is to develop this evaluation system using only foundational C# concepts: choosing appropriate data types, handling precision and conversions, calculating arithmetic expressions, formatting output strings, and organizing conditional logic using decision branches.

Program Requirements

The program must collect the following information from the customer:

- **Physical Weight:** The measured parcel weight in kilograms (may contain decimals).
- **Package Dimensions:** The length, width, and height of the parcel in centimeters (whole numbers).
- **Destination Zone:** The destination region code:
  - 1 = Domestic
  - 2 = Regional Asia-Pacific
  - 3 = International Long-Haul
- **Declared Value:** The total commercial value of the contents in Philippine Pesos (money).
- **Cargo Classification:** The cargo category (general merchandise, perishable goods, hazardous materials, or high-value electronics).
- **Commercial Customs Invoice:** Whether the package includes an official commercial invoice (yes or no).
- **Shipping Insurance:** Whether the sender requested transit insurance coverage (yes or no).

Assessment and Calculation Rules

Implement the following rules strictly in the order described.

1. Volumetric Weight and Chargeable Weight

Freight carriers calculate costs using whichever weight is larger: the physical weight or the space the package occupies in the cargo hold.

- The volume of the parcel is calculated by multiplying its length, width, and height in centimeters.
- The volumetric weight in kilograms is calculated by dividing the total volume by 5,000. Ensure your division does not truncate decimal points.
- The chargeable weight is determined by comparing the physical weight and the volumetric weight. Whichever is greater becomes the chargeable weight.

2. Physical Limits and Safety Restrictions

- Any package where any single dimension (length, width, or height) exceeds 150 cm, or where the total volume exceeds 300,000 cubic centimeters, is automatically **SHIPMENT REJECTED** for exceeding conveyor belt limits.
- The parcel's physical weight must be greater than zero and cannot exceed 70 kilograms. Any package outside this range is automatically **SHIPMENT REJECTED**.
- Hazardous cargo restrictions:
  - Hazardous materials destined for International Long-Haul (Zone 3) are automatically **SHIPMENT REJECTED** due to commercial airline safety regulations.

    - Hazardous materials destined for Domestic (Zone 1) or Regional (Zone 2) must be HELD FOR MANUAL REVIEW to verify dangerous goods handling certification.

3. Customs Compliance and Documentation

- Packages without an official commercial invoice:
    - If the declared value exceeds ₱10,000, the package is automatically SHIPMENT REJECTED for customs non-compliance.
    - If the declared value is ₱10,000 or below, the package must be HELD FOR MANUAL REVIEW for manual declaration processing.
- Packages with a declared value exceeding ₱500,000 are classified as high-liability. If the sender did not purchase shipping insurance, the shipment must be HELD FOR MANUAL REVIEW.

4. Shipping Fee and Tariff Computations

Base Shipping Cost

The base rate per chargeable kilogram is determined by the destination zone:

- **Zone 1 (Domestic):** ₱120.00 per kilogram
- **Zone 2 (Regional Asia-Pacific):** ₱350.00 per kilogram
- **Zone 3 (International Long-Haul):** ₱780.00 per kilogram
- If an invalid zone number is provided, the shipment is immediately SHIPMENT REJECTED.

Calculate the base shipping cost by multiplying the chargeable weight by the rate for that zone.

Cargo Surcharges

Additional handling fees are applied based on the cargo classification:

- **Perishable goods:** Add a 25% surcharge to the base shipping cost.
- **Hazardous materials:** Add a 50% surcharge to the base shipping cost.
- **High-value electronics:** Add a fixed security handling surcharge of ₱1,500.00.
- **General merchandise:** No surcharge (₱0.00).

Transit Insurance Fee

- If insurance was requested: The fee is 1.5% of the declared value, with a minimum required fee of ₱500.00 (whichever is greater).
- If insurance was not requested: The insurance fee is ₱0.00.

Customs Duties and Taxes

- **Domestic shipments (Zone 1):** Completely exempt from customs duties (₱0.00).
- **Regional and International shipments (Zone 2 and Zone 3):**
    - Declared value up to ₱10,000.00: Duty-free under the de minimis threshold (₱0.00 duty).
    - Declared value between ₱10,000.01 and ₱100,000.00: 5% customs duty on the declared value.
    - Declared value exceeding ₱100,000.00: 12% customs duty on the declared value.

Total Amount Due

Calculate the final total billable charge by summing the base shipping cost, cargo surcharge, transit insurance fee, and customs duty.

5. Final Operational Decision Hierarchy

> **Strict Priority Hierarchy:**
> **SHIPMENT REJECTED** > **HELD FOR MANUAL REVIEW** > **CLEARED FOR DISPATCH**
> If any rule flags the package for rejection, the final decision is SHIPMENT REJECTED. If no rejection occurred, but one or more rules flagged the package for inspection, the final decision is HELD FOR MANUAL REVIEW. If the package meets all acceptance rules and requires no special intervention, the final decision is CLEARED FOR DISPATCH.

Output Display Requirements

The program must display a structured summary report in the console with formatted numbers, aligned columns, and currency amounts displayed to two decimal places:

```
============================================================
           CARGO DISPATCH & CUSTOMS INVOICE
============================================================
Package Metrics:
  Actual Weight      : 12.50 kg
  Dimensions         : 40 x 30 x 25 cm (Volumetric: 6.00 kg)
  Chargeable Weight  : 12.50 kg
  Destination        : Zone 2 (Regional)
  Classification     : High-Value Electronics

Financial Breakdown:
  Declared Value     : PHP 120,000.00
  Base Shipping Fee  : PHP 4,375.00
  Surcharge Applied  : PHP 1,500.00
  Insurance Cost     : PHP 1,800.00
  Customs Duty       : PHP 14,400.00
  ------------------------------------------------------------
  TOTAL AMOUNT DUE   : PHP 22,075.00

Operational Status:
  Decision           : CLEARED FOR DISPATCH
============================================================
```

Sample Test Scenarios for Verification

Scenario	Given Customer Information	Expected Values	Expected Status
Scenario A: Standard Regional Shipment	Weight: 8.0 kg Dimensions: 30 × 20 × 20 cm Destination: Zone 2 (Regional) Classification: General merchandise Value: ₱8,500.00 Invoice: Yes, Insurance: No	Volumetric: 2.40 kg Chargeable: 8.00 kg Base Shipping: ₱2,800.00 Duty: ₱0.00 (Exempt) Total: ₱2,800.00	CLEARED FOR DISPATCH
Scenario B: Hazardous Cargo Long-Haul	Weight: 15.0 kg Dimensions: 50 × 40 × 30 cm Destination: Zone 3 (International) Classification: Hazardous materials Value: ₱25,000.00 Invoice: Yes, Insurance: Yes	Chargeable: 15.00 kg Hazardous materials prohibited on Zone 3 passenger routes.	SHIPMENT REJECTED
Scenario C: Low-Value Domestic Missing Invoice	Weight: 3.0 kg Dimensions: 20 × 15 × 10 cm Destination: Zone 1 (Domestic) Classification: General merchandise Value: ₱4,000.00 Invoice: No, Insurance: No	Chargeable: 3.00 kg Missing commercial invoice for parcel valued at or below ₱10,000.	HELD FOR MANUAL REVIEW